Learning Objective 9.1: I understand the process of decoupling and finding solutions for Maxwell's equations for dynamic fields.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **A wave in a lossless dielectric.** A 1 GHz wave travels in the $+z$ direction in a lossless dielectric with $\epsilon_r = 9$ and $\mu_r = 1$. Its electric field is along $\hat{\mathbf{x}}$, with amplitude 10 V/m and phase zero at $z = 0$.

   (a) Find β, λ, and v_p.

$$\beta = \frac{\omega\sqrt{\epsilon_r}}{c} = \frac{2\pi \times 10^9 \times 3}{2.998 \times 10^8} = 62.9 \text{ rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = 9.99 \text{ cm}, \qquad v_p = \frac{c}{3} = 1.00 \times 10^8 \text{ m/s}$$

β =	62.9 rad/m
λ =	9.99 cm
v_p =	1.00×10^8 m/s

   (b) Write the phasor $\widetilde{\mathbf{E}}(z)$ and the time-domain field $\mathbf{E}(z,t)$.

Only the forward wave is present, so $E_0^- = 0$ and $E_0^+ = 10$ V/m:

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\, 10 e^{-j62.9z} \text{ V/m}$$

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\, 10 \cos\left(2\pi \times 10^9 t - 62.9z\right) \text{ V/m}$$

2. **The exact attenuation constant.**

(a) Starting from $\gamma^2 = j\omega\mu\,(\sigma + j\omega\epsilon)$ with $\gamma = \alpha + j\beta$, derive the two equations $\alpha^2 - \beta^2 = -\omega^2\mu\epsilon$ and $2\alpha\beta = \omega\mu\sigma$, and from them the exact expression for α.

$$(\alpha + j\beta)^2 = \alpha^2 - \beta^2 + j2\alpha\beta = -\omega^2\mu\epsilon + j\omega\mu\sigma$$

Matching real and imaginary parts gives the two equations. The second gives $\beta = \omega\mu\sigma/2\alpha$; substituting into the first and multiplying by α^2,

$$\left(\alpha^2\right)^2 + \omega^2\mu\epsilon\,\alpha^2 - \frac{\omega^2\mu^2\sigma^2}{4} = 0$$

$$\alpha^2 = \frac{\omega^2\mu\epsilon}{2}\left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1\right]$$

We keep the root with $\alpha^2 > 0$, and α is its positive square root.

(b) Wet earth has $\sigma = 0.01$ S/m and $\epsilon_r = 10$. Find $\sigma/\omega\epsilon$ at 1 MHz and at 1 GHz, and say which limit, if either, applies at each.

$$1\text{ MHz}: \quad \frac{\sigma}{\omega\epsilon} = \frac{0.01}{2\pi \times 10^6 \times 10 \times 8.854 \times 10^{-12}} = 18.0$$

$$1\text{ GHz}: \quad \frac{\sigma}{\omega\epsilon} = 0.0180$$

At 1 MHz the earth is close to a good conductor; at 1 GHz it is a low-loss dielectric.

(c) Find α at each frequency from the matching limit, and compare with the exact values, 0.193 Np/m at 1 MHz and 0.596 Np/m at 1 GHz.

At 1 MHz, the good-conductor limit:

$$\alpha \approx \sqrt{\pi f \mu\sigma}$$

$$= \sqrt{\pi \times 10^6 \times 4\pi \times 10^{-7} \times 0.01} = 0.199\text{ Np/m}$$

At 1 GHz, the low-loss limit:

$$\alpha \approx \frac{\sigma}{2}\sqrt{\frac{\mu_0}{\epsilon}} = 0.005 \times \frac{376.7}{\sqrt{10}} = 0.596\text{ Np/m}$$

The good-conductor value is 3% high, because a ratio of 18 is not yet far above one; the low-loss value matches the exact one to three figures.

α, 1 MHz =	0.199 Np/m
α, 1 GHz =	0.596 Np/m

3. **Skin depth.** Copper has $\sigma = 5.8 \times 10^7$ S/m and aluminum 3.7×10^7 S/m; both have $\mu = \mu_0$.

(a) Find the skin depth of copper at 60 Hz, 1 MHz, and 10 GHz.

$$\delta = \frac{1}{\sqrt{\pi f \mu_0 \sigma}}$$

$$60\text{ Hz}: \ \delta = 8.53\text{ mm}$$

$$1\text{ MHz}: \ \delta = 66.1\ \mu\text{m}$$

$$10\text{ GHz}: \ \delta = 0.661\ \mu\text{m}$$

Each factor of 100 in frequency divides δ by 10.

60 Hz = 8.53 mm

1 MHz = $66.1\ \mu$m

10 GHz = $0.661\ \mu$m

(b) A shield of aluminum foil $16\ \mu$m thick surrounds a circuit. At 100 MHz, what fraction of the field amplitude that enters the foil reaches its far side? Express the reduction in dB.

$$\delta = \frac{1}{\sqrt{\pi \times 10^8 \times 4\pi \times 10^{-7} \times 3.7 \times 10^7}} = 8.27\ \mu\text{m}$$

$$e^{-t/\delta} = e^{-16/8.27} = e^{-1.93} = 0.145$$

$$8.686 \times 1.93 = 16.8\text{ dB}$$

Lesson 38 adds the reflection at the foil's surface, which reduces the field further.

fraction = 0.145

reduction = 16.8 dB

(c) Seawater has $\sigma = 4$ S/m and $\epsilon_r = 81$. At 10 kHz, where it is a good conductor, find the depth at which the field amplitude falls to 1% of its value at the surface.

$$\delta = \frac{1}{\sqrt{\pi \times 10^4 \times 4\pi \times 10^{-7} \times 4}} = 2.52\text{ m}$$

$$e^{-d/\delta} = 0.01$$

$$d = \delta \ln 100 = 4.61 \times 2.52 = 11.6\text{ m}$$

depth = 11.6 m

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